Introduction

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Nearly all of us have to work during our lives. We reach adulthood and seek employment, and then we work for the next 30 to 50 years. Work is our primary means of livelihood. It serves an important economic function in the global world of commerce. It creates opportunities for social interactions and friendships. And it provides the products and services that sustain and improve our standard of living.

This text is all about work and the systems by which it is accomplished. The text also examines the principles and programs that allow work to be performed most efficiently and safely, and it discusses the techniques used to measure and manage work. The common denominator in the analysis, design, and measurement of work is time. For many reasons that are enumerated in this chapter time is important in work. In general, it is desirable to accomplish a given task in the shortest possible time.

Work systems constitute the central theme in the discipline of industrial engineering. Each engineering discipline is concerned with its own type of technical system. The correlations are listed in Table 1 for the major engineering disciplines. In most cases, each engineering field has expanded to include several related kinds of systems, and these additional systems are also listed in the table. Their addition has been a natural evolutionary process that has paralleled the development of new technologies. For industrial engineering, the additional subjects include operations research, ergonomics,

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discrete event simulation, and information systems. We can reasonably argue that all of these fields are related to the study of systems that perform work.

In this chapter, we define (1) work itself, (2) work systems, (3) jobs and occupations, and (4) productivity. In Section 5, we discuss the variety of topics covered in this text and the way in which the topics are organized. All of these topics are concerned with work in one way or another. Some familiar quotations and sayings about work and jobs are presented in Table 2 for the reader's amusement. Work has been the object

TABLE 1 Engineering Disciplines and the Types of Systems They Design and Analyze

Engineering Discipline	Type of System	New Systems and Subjects
Aeronautical engineering	Airplanes	Aerospace systems, missile systems
Chemical engineering	Chemical systems, chemical processing systems	Processing of integrated circuits, process control, polymer science, biotechnology
Civil engineering	Structural systems (e.g., bridges, buildings, roads)	Environmental systems, transportation systems, fluid systems
Electrical engineering	Electrical systems, power generation systems	Computer systems, integrated circuits, control systems
Industrial engineering	Work systems	Operations research, ergonomics, simulation, information systems
Mechanical engineering	Mechanical systems, thermal systems	Control systems, computer-aided design, micro-electro-mechanical systems
Metallurgical engineering[a]	Metals systems	Ceramics, polymers, composites, electron microscopy

[a]The name *materials science and engineering* has largely replaced *metallurgical engineering*.

TABLE 2 Notable Observations About Work and Jobs

I do not like work even when someone else does it. (Mark Twain)
All work and no play make Jack a dull boy—to everyone but his employer.
Man may work from sun to sun, but a woman's work is never done.
A woman's work is never done, especially the part she asks her husband to do.
The people who claim that brain work is harder than physical work are generally brain workers.
Many thousands of people are already working a four-day week; the trouble is it takes them five days to do it.
Hard work never hurt anyone who hired someone else to do it.
Hard work pays off in the future; laziness pays off now.
Some people are so eager for success that they are even willing to work for it.
It isn't the hours you put into your work that count, it's the work you put into the hours.
What will happen to work when the trend toward longer education meets the trend toward earlier retirement?
Genius is one percent inspiration and ninety-nine percent perspiration. (Thomas A. Edison)
The difference between a job and a career is the difference between 40 and 60 hours a week. (Robert Frost)
The softer the job, the harder it is to get.
The best man for the job is often a woman.
Two can live as cheaply as one—if they both have good jobs.

Source: Compiled from J. Bartlett, *Familiar Quotations*, 14th ed., E. M. Beck, ed. (Boston: Little, Brown, 1968); E. Esar, *20,000 Quips and Quotes*, (New York: Barnes and Noble, 1995); and other sources.

of study for many years, and some of the more significant findings and personalities that have contributed to this field are described in Historical Note 1.

HISTORICAL NOTE 1 THE STUDY OF WORK

There is evidence that the study of work and some of the basic principles about work originated in ancient times [1]. The Babylonians used the principle of a ***minimum wage*** around 1950 B.C. The Chinese organized work according to the principle of ***labor specialization*** around 1644 B.C. The ancient Romans used a primitive form of ***factory system*** for the production of armaments, textiles, and pottery. They also perfected the military organizational structure, which is the basis for today's ***line and staff organization*** of work.

The ***Industrial Revolution*** started in England around 1770 with the invention of several new machines used in the production of textiles,[1] ***James Watt***'s steam engine, and ***Henry Maudslay***'s screw-cutting lathe. These inventions resulted in fundamental changes in the way work was organized and accomplished: (1) the transfer of skill from workers to machines, (2) the start of the machine tool industry, based on Maudslay's lathe and other new types of metal-cutting machines that allowed parts to be produced more quickly and accurately than the prior handicraft methods, and (3) the introduction of the factory system in textile production and other industries, which used large numbers of unskilled workers (including women and children) who labored long hours for low pay. The factory system employed the specialization of labor principle.

While England was leading the Industrial Revolution, the important concept of ***interchangeable parts manufacture*** was being introduced in the United States. Much credit for this concept is given to ***Eli Whitney*** (1765–1825), although others had recognized its importance [7]. In 1797, Whitney negotiated a contract to produce 10,000 muskets for the U.S. government. The traditional way of making guns at the time was to custom-fabricate each part for a particular gun and then hand-fit the parts together by filing. Each musket was therefore unique. Whitney believed that the components could be made accurately enough to permit parts assembly without fitting. After several years of development in his Connecticut factory, he was able to demonstrate the principle before government officials, including Thomas Jefferson. His achievement was made possible by the special machine tools, fixtures, and gauges that he had developed. Interchangeable parts manufacture required many years of refinement in the early and mid-1800s before becoming a practical reality, but it revolutionized work methods used in manufacturing. It is a prerequisite for mass production of assembled products.

The mid- and late 1800s and early 1900s witnessed the introduction of several consumer products, including the sewing machine, bicycle, and automobile. In order to meet the mass demand for these products, more efficient production methods were required. Some historians identify developments during this period as the Second

[1]The machines included (1) James Hargreaves's Spinning Jenny, patented in 1770; (2) Richard Arkwright's Water Frame, developed in 1771; (3) Samuel Crompton's Mule-Spinner, developed around 1779; and (4) Edmund Cartwright's Power Loom, patented in 1785. Historians sometimes include James Kay's Flying Shuttle introduced in 1733 in the list of great inventions of the Industrial Revolution.

Industrial Revolution, characterized in terms of its effects on work systems by the following: (1) mass production, (2) assembly lines, and (3) scientific management.

Mass production was primarily an American phenomenon. Its motivation was the mass market that existed in the United States, where the population in 1900 was 76 million and growing and by 1920 exceeded 106 million. Such a large population, larger than any western European country, created a demand for large numbers of products. Mass production provided those products. Certainly one of the important technologies of mass production was the moving ***assembly line***, introduced by ***Henry Ford*** (1863–1947) in 1913 at his Highland Park plant. The assembly line was a new form of work system and made possible the mass production of complex consumer products. Use of assembly line methods permitted Ford to sell a Model T automobile for less than $500 in 1916, thus making ownership of cars feasible for a large segment of the American population.

The ***scientific management*** movement started in the late 1800s in the United States in response to the need to plan and control the activities of growing numbers of production workers. The most important members of this movement were Frederick W. Taylor, Frank Gilbreth, and Lillian Gilbreth. The principal approaches of scientific management were the following: (1) motion study, aimed at finding the best method to perform a given task and eliminating delays; (2) time study to establish work standards for a job; (3) extensive use of standards in industry; (4) the piece rate system and similar labor incentive plans; and (5) use of data collection, record keeping, and cost accounting in factory operations. These approaches, while revolutionary at the time they were first implemented, are fundamental and indispensable techniques used today in business and industry for work management.

Frederick W. Taylor (1856–1915) is known as the "father of scientific management" for his application of systematic approaches to the study and improvement of work. His findings and writings have influenced factory management in virtually every industrialized country in the world, especially the United States. It can be argued that much of the mass production power of the United States in the twentieth century is due to the scientific management principles espoused by Taylor.

Born in Philadelphia into an upper middle class family, Taylor attended the local Germantown Academy and then Phillips Exeter Academy in New Hampshire. His father's wish was for young Taylor to pursue a legal career. However, after passing the entrance tests for Harvard University, poor eyesight forced him to abandon those plans.[2] Instead, he became an apprentice patternmaker and machinist in 1874. In 1878, he became an employee at the Midvale Steel Company in Philadelphia and during the next 12 years progressed from machine shop worker to gang boss, foreman, and master mechanic. In 1883, he earned a mechanical engineering degree from Stevens Institute of Technology through part-time study and was promoted to chief engineer at the company.

[2]Taylor's poor eyesight was apparently caused by too much nighttime study at Phillips Exeter. He was at the top of his class at the Academy. There is some speculation that young Taylor did not want to follow his father's profession as a lawyer. He preferred to follow his own career path. His eyesight was largely recovered by 1875.

Taylor introduced time study at Midvale Steel sometime between 1881 and 1883 while he was foreman of the machine shop. By 1883, a given task had been divided into work elements, and the timing of each element and then summing the times had been found to be more useful than timing the whole task. Taylor's belief was that by studying each element, wasted motions could be eliminated and efficiency could thereby be improved in every step of an operation. While at Midvale, Taylor also conducted metal-cutting experiments on the company's products, which included military cannons and locomotive wheels. The experiments resulted in significant improvements in productivity in the plant. However, not all of Taylor's proposals about work were successful. A notable example was his idea to separate the shop foreman's job into eight specialized functions. The resulting structure is called a functional organization.

Taylor left Midvale Steel in 1889. From 1890 to 1893, he served as general manager of a company that processed wood pulp. He then became a management consultant from 1893 to 1901. During this latter period, his most important engagement was with the Bethlehem Iron Works (predecessor to the Bethlehem Steel Company) between 1898 and 1901. Two famous experiments were conducted at Bethlehem: (1) the shoveling experiment and (2) pig iron handling.

In the ***shoveling experiment***, Taylor observed that each yard worker brought his own shovel to work, and the shovels were all different sizes. The workers were required to shovel various materials in the yard, such as ashes, coal, and iron ore. Because the densities of these materials differed, it meant that the weight per shovelful varied significantly. Through experimentation with the workers, he determined that different-sized shovels should be used for the different materials. His conclusion was that the appropriate shovel size was one in which the load was 21 pounds. This load weight maximized the amount of work that could be accomplished each day by a worker and minimized the costs to the company.

In the study of ***pig iron handling***,[3] Taylor believed that the yard workers who loaded pig iron from the storage yard into freight cars were not using the best method. The workers seemed to work too hard and then had to rest for too long to recover from the exertion. Their daily wage was $1.15 (in 1898) and they averaged 12.5 tons per day. Taylor confronted one of the men named Schmidt and offered him the opportunity to earn $1.85 per day if he followed Taylor's instructions on how to perform the work. The instruction consisted of improvements in the way the pig iron was picked up, carried, and dropped off, combined with more frequent but shorter rest breaks. Enticed by the opportunity to earn more money, Schmidt agreed.[4] By using the improved method, Schmidt was able to consistently load 47 tons per day. Other workers were eager to sign on for the higher pay.

[3]Pig iron is the iron tapped from a blast furnace. It contains impurities and must be subsequently refined to make cast iron and steel.

[4]In fact, Taylor dreamed up the name Schmidt. The worker's real name was Henry Knolle. Opponents of Taylor and scientific management circulated reports in 1910 that Schmidt had died from overworking to achieve the 47 tons per day. The truth is that Henry Knolle lived on until 1925, dying at the age of 54.

In 1901, Taylor retired to his estate in Philadelphia but continued to promote the emerging field of scientific management through lectures and publications such as ***Shop Management*** (1903), ***On the Art of Cutting Metals*** (1906), and ***Principles of Scientific Management*** (1911). Perhaps Frederick W. Taylor's most important contribution in the improvement of work was his successful promotion and promulgation of the field of scientific management during the period of his retirement.

Frank Gilbreth (1868–1924) is noted for his pioneering efforts in analyzing and simplifying manual work. He was associated with the scientific management movement in the late 1800s and early 1900s, in particular for his achievements in motion study. He is sometimes referred to as the "father of motion study." Two of his important theories about work were (1) that all work was composed of 17 basic motion elements that he called "therbligs" and (2) the principle that there is "one best method" to perform a given task.

As a young man, Gilbreth had planned to attend college but was forced to get a job due to his father's untimely death. He started as a bricklayer's apprentice in 1885 at age 17. On his first day at work, Gilbreth noticed that the bricklayer assigned to teach him laid bricks in three different ways, one in normal working, a second when working fast, and a third when instructing Gilbreth. He also noticed that other bricklayers used other methods, all of which seemed to be different. It occurred to Gilbreth that there should be one best way to accomplish bricklaying, and all bricklayers should use that one best method. Gilbreth analyzed the work elements that were required in bricklaying, attempting to simplify the task and eliminate wasted motions. He was able to develop a method that reduced the number of steps required to lay one brick by about 70 percent. He participated in the development of a bricklayer's scaffold that could be adjusted in height so the worker would have bricks and mortar at the same elevation level as he was working rather than be required to stoop to ground level to retrieve these materials. The movable scaffold is still used today.

By age 26, Gilbreth decided to go into business for himself and became a widely recognized building contractor in New York City. One of the reasons for his success was the capability of his crews to complete a project quickly, a result of his interest in time and motion study. He was able to apply the principles of time and motion study to the labor of construction workers and other industrial employees so as to increase their efficiency and output. Ultimately, he would become one of the leading public speakers for Taylor's scientific management movement.

In 1904, Gilbreth married Lillian Moller, a teacher and psychologist who collaborated in his research on motion study, contributing an emphasis on the human and social attributes of work. In 1911, Gilbreth published ***Motion Study***, a book that documented many of their research findings. Around 1915, they founded the management consulting firm of Frank B. Gilbreth, Inc. When Frank died prematurely in 1924, Lillian assumed the presidency of the firm and carried on his work as a researcher and advocate of motion and time study. She became a noted educator, author, engineer, and consultant in her own right.

Lillian Gilbreth (1878–1972) was the oldest of nine children. She earned bachelor's and master's degrees at the University of California, Berkeley. During her marriage to Frank, she became the mother of 12 children and earned a doctorate at Brown

University in 1915, a rare achievement for a woman at the time. Her other accomplishments were equally noteworthy. With her husband, she co-authored four books: ***A Primer of Scientific Management*** (1914), ***Fatigue Study*** (1916), ***Applied Motion Study*** (1917), and ***Motion Study for the Handicapped*** (1917). On her own, she wrote ***The Psychology of Management*** (1914) and several other books after Frank's death. She also held faculty positions at Purdue University (1935–1948) and several other universities. Among the many honors and awards she received during her lifetime, one of the most significant was her election to the National Academy of Engineering. She was the first woman ever to be elected to the NAE. With good reason, she has been called the "First Lady of Engineering."

The story of how Frank and Lillian Gilbreth practiced efficiency and motion study in their own home was humorously documented by two of their 12 children in 1949 with the publication of ***Cheaper by the Dozen***. It was made into a motion picture in 1950.

1 THE NATURE OF WORK

For our purposes, ***work*** is defined as an activity in which a person exerts physical and mental effort to accomplish a given task or perform a duty. The task or duty has some useful objective. It may involve one or more steps in making a product or delivering a service. The worker performing the task must apply certain skills and knowledge to complete the task or duty successfully. There is usually a commercial value in the work activity, and the worker is compensated for performing it. By commercial value, we mean that the task or duty contributes to the buying and selling of something (e.g., a product or service), which ultimately provides the means of paying for the work. Work is also performed in government, but its value is surely measured on a scale other than commercial.

In physics, ***work*** is defined as the displacement (distance) that an object moves in a certain direction multiplied by the force acting on the object in the same direction. Thus, physical work is measured in units of newton-meters (N-m) in the International System of Units (metric system) or foot-pounds (ft-lb) in U.S. customary units. This definition can be reconciled with our labor-oriented work definition by imagining a material handling worker pushing a cart across a warehouse floor by exerting a force against the cart to move it a certain distance. However, human work includes many activities other than the muscular application of forces to move objects. Nearly all human work activities include both physical and mental exertions by the worker. In some cases, the physical component dominates the activity, while in others the mental component is more important. In virtually all work situations, the activity cannot be performed unless the worker applies some combination of physical and cognitive effort.

1.1 The Pyramidal Structure of Work

Work consists of tasks. A ***task*** is an amount of work that is assigned to a worker or for which a worker is responsible. The task can be repetitive (as in a repetitive operation in mass production) or nonrepetitive (performed periodically, infrequently, or only once). A task can be divided into its constituent activities, which form the pyramidal structure

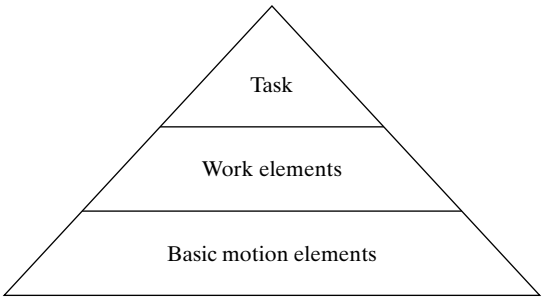


Figure 1 The pyramidal structure of a task. Each task consists of multiple work elements, which in turn consist of multiple basic motion elements.

shown in Figure 1. Each task consists of several work elements, and each work element consists of basic motion elements. ***Basic motion elements*** are actuations of the limbs and other body parts while engaged in performing the task. These basic motion elements include reaching for an object, grasping an object, or moving an object. Other basic motions include walking and eye movement (e.g., eye focusing, reading). Multiple basic motion elements are generally required to perform a ***work element***, which is defined as a series of work activities that are logically grouped together because they have a unified function within the task. For example, a typical assembly work element consists of reaching for a part, grasping it, and attaching it to a base part, perhaps using one or more fasteners (e.g., screws, bolts, and nuts). Many such work elements make up the total work content of assembling all of the components to the base part. Work elements usually take six seconds or longer, while a basic motion element may take less than a second. The entire task may take 30 seconds to several minutes if it is a repetitive task, while nonrepetitive tasks may require a much longer time to complete.

Just as a task can be divided into its component activities (work elements each consisting of multiple basic motion elements), the typical job of a worker is likely to consist of more than one task. Thus the job adds a next higher level to the existing pyramid. Furthermore, a worker's career is likely to consist of more than one job, as he or she changes employers and/or advances through several jobs with a single employer during a lifetime of working. Accordingly, one might envision the ultimate work pyramid as consisting of the five levels shown in Figure 2.

1.2 Importance of Time

In nearly all human endeavors, "time is of the essence." In sports, time is often the major factor in deciding the outcome of a contest. The fastest time wins the race. In games that use time periods, the victor must score the winning points within the limits of those periods. In other aspects of life, time is also important. Students must be in class on time. They must complete an hour quiz in 50 minutes (or other designated time limit). Medical patients must schedule appointments with their doctors at specific times. When we drive to a given

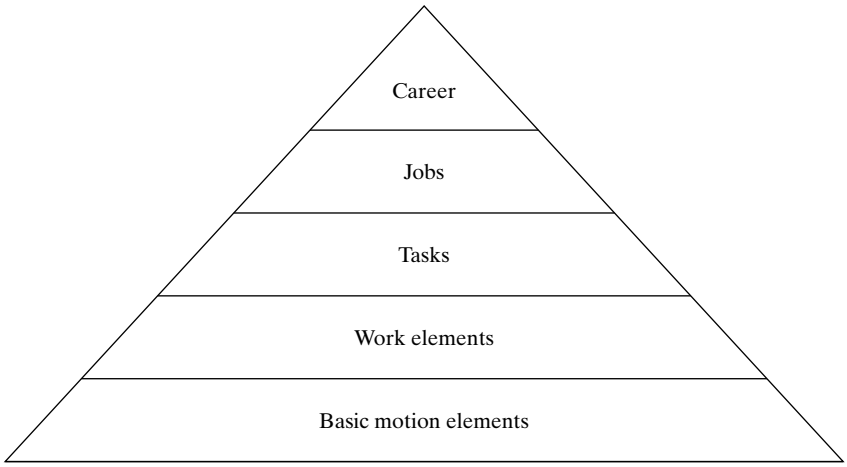


Figure 2 The pyramidal structure of work.

destination, whether it is a long vacation trip or a nearby shopping expedition, we select the route that will get us there in the shortest possible time. Why waste time and gasoline?

Time is important in business and industry, as the following examples demonstrate:

- *New product introduction.* The manufacturer introducing a new product to the market in the shortest time is usually the one rewarded with the most profits.
- *Product cost.* In many cases, the number of labor hours required to produce a product represents a significant portion of total manufacturing cost, which determines the price of the product. Companies that can reduce the time to make a product can sell it at a more competitive price.
- *Delivery time.* Along with cost and quality, delivery time is a key criterion in vendor selection by many companies. The supplier that can deliver its products in the shortest time is often the one selected by the customer.
- *Overnight delivery.* The success of overnight delivery offered by parcel transport companies (e.g., UPS, Fed Ex) illustrates the growing commercial importance of time.
- *Competitive bidding.* In many competitive bidding situations, proposals must be submitted by a specified date and time. Late proposals will be disregarded.
- *Production scheduling.* The production schedule in a manufacturing plant is based on dates and times.

Time is important in work. The many reasons why this is so include the following:

- The most frequently used measure of work is time. How many minutes or hours are required to perform a given task?
- Most workers are paid according to the amount of time they work. They earn an hourly wage rate or a salary that is paid on a weekly, biweekly, or monthly basis.
- Workers must arrive at work on time. If a worker is a member of a work team, his or her absence or tardiness may handicap the rest of the team.

- When production workers are paid on an incentive plan, they earn their bonuses based on how much time they can save relative to the standard time for a given task.
- Labor and staffing requirements are computed using workloads measured in units of time. For example, how many workers will be required during a 40-hour week to accomplish a workload of 600 hours?

2 DEFINING WORK SYSTEMS

A work system can be defined as (1) a physical entity and (2) a field of professional practice. Both definitions are useful in studying the way work is accomplished.

2.1 Physical Work Systems

As a physical entity, a ***work system*** is a system consisting of humans, information, and equipment that is designed to perform useful work, as illustrated in Figure 3. The result of the useful work is a contribution to the production of a product or the delivery of a service. Examples of work systems include the following:

- A worker operating a production machine in a factory
- An assembly line consisting of a dozen workers at separate workstations along a moving conveyor
- A robotic spot welding line in an automobile final assembly plant performing spot welding operations on sheet metal car bodies
- A freight train transporting 60 intermodal container cars from Los Angeles to Chicago
- A parcel service agent driving a delivery truck to make customer deliveries in a local area
- A receptionist in an office directing visitors to personnel in the office and answering incoming telephone calls

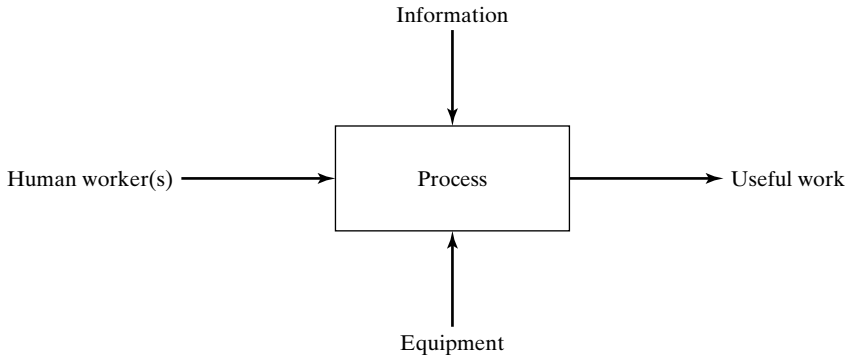


Figure 3 A work system consists of human workers, information, and equipment designed to accomplish useful work by means of a process.

- A designer working at a computer-aided design (CAD) station to design a new product
- A construction project consisting of a work crew building a highway bridge

As our list indicates, a work system can include one or more human workers. It can also include automated systems that operate for extended periods of time without human attention. Sooner or later, automated systems require the attention of human workers for purposes of maintenance or reprogramming or other reasons. The information associated with the work system may consist simply of a worker's knowledge required to perform an operation, or it may involve the use of databases and programs that must be accessed using computer systems. The equipment in a work system may be a single machine tool operated by one worker or a collection of automated machines that operate under computer control in a coordinated fashion.

2.2 Work Systems as a Field of Professional Practice

As a field of professional practice, ***work systems*** includes (1) work methods, (2) work measurement, and (3) work management.[5] The term ***work science*** is often used for this professional practice. The field of ***work methods*** consists of the analysis and design of tasks and jobs involving human work activity. Terms related to work methods include ***operations analysis*** and ***methods engineering***. The term ***motion study*** is also used, but its scope is usually limited to the physical motions, tools, and workplace layout used by a worker to perform a task. The other terms are less restrictive and include the analysis and design of complex processes consisting of material and information flows through multiple operations.

Work measurement is the analysis of a task to determine the time that should be allowed for a qualified worker to perform the task. The time thus determined is called the ***standard time***. Among its many applications, the standard time can be used to compute product costs, assess worker performance, and determine worker requirements (e.g., how many workers are needed to accomplish a given workload). Because of its emphasis on time, work measurement is often referred to as time study. However, in the modern context, ***time study*** has a broader meaning that includes work situations in which an operation is performed by automated equipment, and it is desired to determine the cycle time for the operation. Thus, time study covers any and all work situations in which it is necessary to determine how long it takes to accomplish a given unit of work, whether the work unit is concerned with the production of a product or the delivery of a service. Time is important because it equates to money ("time is money," as the saying goes), and money is a limited resource that must be well managed in any organization.

Work management refers to the various organizational and administrative functions that must be accomplished to achieve high productivity of the work system and effective supervision of workers. Work management includes functions such as (1) organizing workers to perform the specialized tasks that constitute the workload in each

[5]The title of this text emphasizes the two definitions: (1) work systems as physical entities and (2) work systems as a field of professional practice.

department of the company or other organization, (2) motivating workers to perform the tasks, (3) evaluating the jobs in the organization so that each worker is paid an appropriate wage or salary commensurate with the type of work performed, (4) appraising the performance of workers to reward better-performing workers appropriately, and (5) compensating workers using a rational payment system for the work they perform.

3 TYPES OF OCCUPATIONS

The Bureau of Labor Statistics of the U.S. Department of Labor identifies 821 occupations in its Standard Occupational Classification (SOC). The SOC covers virtually every type of work performed for pay or profit in the United States, and organizes these occupations into 23 major groups, based on type of work and/or industry sector. The categories are listed in Table 3. Representative positions and job titles are also given.

Rather than use the 23 categories in the SOC system, it is convenient for our purposes to group occupations into the following four broad categories that reflect the work content and function of the jobs rather than type of work and industry sector:

- *Production workers*, in which the work involves making products.
- *Logistics workers*, in which the work involves moving materials, products, or people.
- *Service workers*, in which the work involves one or more of the following functions: (1) providing a service, (2) applying existing information and knowledge, and (3) communicating.
- *Knowledge workers*, in which the work involves one or more of the following functions: (1) creating new knowledge, (2) solving problems, and (3) managing.

These categories are correlated with the types of work systems used in business, industry, and government. Production workers have jobs in production systems (manufacturing, construction), logistics workers are employed in logistics systems (transportation, material handling), service workers work in service operations (retail, government, health care), and knowledge workers are involved in knowledge-oriented activities (creative work, problem solving, management).

The characteristics and representative job titles of the four categories are summarized in Table 4. We see that production and logistics workers engage in mostly physical labor, while service and knowledge workers perform duties that require mostly cognitive activities. Overlaps exist. Production and logistics workers must utilize their brains in their jobs, while the work of service and knowledge workers usually includes physical activities.

A key feature among the four categories is worker discretion [17], which refers to the necessity to make responsible decisions and exercise judgment in carrying out the duties of the position. Jobs that are highly standardized and routine require minimum worker discretion, while jobs in which workers must adapt their behavior in response to variations in the work situation require high discretion. As indicated in Table 4, production and logistics workers are required to use limited or moderate discretion in performing their duties, while service and knowledge workers must exercise much greater discretion in their jobs. In production and logistics operations, the objective is to complete

TABLE 3 Major Occupation Categories of the U.S. Department of Labor Standard Occupational Classification

SOC #	Occupation Category	Representative Positions and Job Titles
11	Management Occupations	Chief executive officer, construction manager, funeral director, hotel manager, postmaster, plant manager, sales manager, school principal
13	Business and Financial Operations Occupations	Accountant, auditor, claims adjuster, cost estimator, financial analyst, loan officer, purchasing agent, tax collector
15	Computer and Mathematical Occupations	Actuary, computer programmer, systems analyst, database administrator, mathematical analyst, software engineer, statistician
17	Architecture and Engineering Occupations	Architect, drafter, engineering technician, engineers (e.g., civil, electrical, industrial, mechanical), landscape architect, surveyor
19	Life, Physical, and Social Science Occupations	Agricultural scientist, astronomer, chemist, economist, geoscientist, materials scientist, physicist, psychologist, sociologist
21	Community and Social Services Occupations	Clergy, family therapist, marriage counselor, high school counselor, mental health worker, probation officer, social worker
23	Legal occupations	Court reporter, judge, law clerk, lawyer, legal assistant, magistrate, paralegal, title examiner
25	Education, Training, and Library Occupations	College professor, elementary school teacher, high school teacher, kindergarten teacher, librarian, special educator, teacher assistant
27	Arts, Design, Entertainment, Sports and Media Occupations	Actor, artist, athlete, coach, composer, dancer, fashion designer, movie director, musician, news reporter, referee, singer
29	Healthcare Practitioners and Technical Occupations	Chiropractor, clinical technician, dental hygienist, dentist, paramedic, pharmacist, physician, psychiatrist, nurse, therapist, veterinarian
31	Healthcare Support Occupations	Dental assistant, medical assistant, nursing aide, orderly, pharmacy aid, physical therapist, veterinary assistant
33	Protective Services Occupations	Bailiff, detective, firefighter, fire inspector, game warden, jailer, police officer, private investigator, security guard
35	Food Preparation and Serving Occupations	Bartender, chef, cook, dishwasher, dining room host/hostess, fast food worker, food preparation worker, waiter, waitress
37	Building and Grounds Cleaning and Maintenance	Groundskeeper, grounds maintenance worker, housekeeping cleaner, janitor, landscape worker, maid, pest control worker, tree trimmer
39	Personal Care and Service Related Occupations	Animal trainer, barber, bellhop, concierge, cosmetologist, fitness trainer, flight attendant, funeral attendant, hairdresser, usher
41	Sales and Related Occupations	Cashier, salesperson, insurance agent, model, real estate broker, sales engineer, sales representative, telemarketer, travel agent
43	Office and Administrative Support Occupations	Bank teller, bill collector, bookkeeper, hotel desk clerk, office manager, receptionist, police dispatcher, secretary, stock clerk
45	Farming, Fishing, and Forestry Occupations	Agricultural inspector, animal breeder, farm worker, fishing worker, forest and conservation worker, hunter, logging worker, ranch hand
47	Construction and Extraction Occupations	Bricklayer, building inspector, carpenter, cement mason, construction laborer, electrician, oil drill operator, painter, plumber, roofer
49	Installation, Maintenance, and Repair Occupations	Aircraft mechanic, automotive mechanic, home appliance repairer, locksmith, office machine repairer, security system installer
51	Production Occupations	Assembly worker, bookbinder, food processing worker, foundry mold maker, machinist, sewing machine operator, welder
53	Transportation and Material Moving Occupations	Aircraft pilot, bus driver, crane operator, garbage collector, material handling worker, sailor, ship captain, truck driver
55	Military Specific Occupations	Air crew member, infantry soldier, military officer, radar technician, sonar technician, special forces commando

Source: From the Bureau of Labor Statistics, www.bls.gov/soc.

TABLE 4 Comparison of Work Characteristics of Four Categories of Workers

Worker Category	Production Workers	Logistics Workers	Service Workers	Knowledge Workers
Basic functions	Make products	Move materials, products, or people	Provide service Apply information and knowledge Communicate	Create knowledge Solve problems Manage and coordinate
Type of work	Mostly physical	Mostly physical	Mostly cognitive	Mostly cognitive
Worker discretion	Limited	Limited to moderate	Moderate to broad	Broad
Equipment required for basic work	Machinery systems (production tools and machines)	Machinery systems (transportation, material handling)	Computer systems Communication systems	Computer systems Information resources
Industry and professional examples	Manufacturing Construction Agriculture Power generation	Transportation Distribution Material handling Storage	Banking Government service Health care Retail	Management Designing Legal Education Consulting
Representative positions and job titles	Laborer Machine operator Assembly worker Machinist Quality inspector Construction worker	Truck driver Airplane pilot Ship captain Material handler Order picker Shipping clerk	Bank teller Police officer Nurse Physical trainer Salesperson Foreman	Manager Physician Designer Researcher Lawyer Teacher

Source: Adapted from a figure in [17].

TABLE 5 Characteristics of Jobs Requiring Low and High Worker Discretion

Characteristics of Jobs Requiring Low Discretion	Characteristics of Jobs Requiring High Discretion
The work is performed at one location.	Workers determine where they will do their jobs.
Almost any able-bodied person could perform the work if provided with basic training.	The work depends heavily on technical knowledge and prior experience.
The work is dominated by machinery operation, routine procedures, or predetermined activities.	Workers manage their own schedules and the processes used to perform their jobs.
The methods, techniques, and materials are specified.	Workers determine the methods, techniques, and materials they use in their jobs.
The work requires interactions with the same people every day.	Workers must deal with different types of people in their daily work activities.
Work performance is measured primarily in quantitative terms.	Work performance is measured primarily in qualitative terms.

Source: Adapted from [17].

the work precisely as specified and with minimum variation. In service and knowledge operations, the objective is to deal in a discretionary way with situations and people that are inherently variable. Typical characteristics associated with jobs requiring low discretion and jobs requiring high discretion are listed in Table 5.

The relative proportions of workers in various occupational categories are given in Table 6, based on data in an article by Bailey and Barley [2]. The table covers most of the twentieth century for the United States, and indicates trends in the labor force

TABLE 6 Relative Percentages of Occupations in the U.S. Workforce: 1900–1998

Occupational Category	Percentage in 1900	Percentage in 1950	Percentage in 1998	Net Percentage Change 1900–1998
Production and logistics workers:				
Farmworkers	38	12	3	–35
Craft and similar	11	14	11	0
Operatives and laborers	25	26	13	–12
Total production and logistics workers	74[a]	52[a]	27	–47
Service and knowledge workers:				
Service	9	11	16	+7
Sales workers	5	7	11	+6
Clerical and similar	3	12	17	+14
Professional and technical	4	8	18	+14
Managerial and administrative	6	9	11	+5
Total service and knowledge workers	27[a]	47[a]	73	+47

Source: Adapted from [2].
[a]Totals do not add to 100% due to round-off errors.

during this period in which the total U.S. population grew from 76 million to approximately 280 million. The data published in [2] are organized into eight occupational categories, which we have grouped into two major categories, consistent with the worker classification in this text: (1) production and logistics workers and (2) service and knowledge workers. At the beginning of the twentieth century, nearly three-quarters of U.S. workers were employed in farming, skilled trades, factories, and manual labor, which we have grouped as production and logistics workers. Only one-quarter was employed as service and knowledge workers. One hundred years later, the proportions had reversed. Today, most occupations fall within the category of service and knowledge work.

4 PRODUCTIVITY

Productivity is defined as the level of output of a given process relative to the level of input. The term *process* can refer to an individual production or service operation, or it can be used in the context of a national economy. Productivity is an important metric in work systems because improving productivity is the means by which worker compensation can be increased without increasing the costs of the products and services they produce. This leads to more products and services at lower prices for consumers, which improves the standard of living for all. In this section we define labor productivity and examine the theoretical basis for increasing productivity.

4.1 Labor Productivity

The most common productivity measure is ***labor productivity***, defined by the following ratio:

$$LPR = \frac{WU}{LH} \tag{1}$$

where LPR = labor productivity ratio, WU = work units of output, and LH = labor hours of input. The definition of output work units depends on the process under consideration. For example, in the steel industry, tons of steel is the common measure. In the automobile industry, the number of cars produced is the appropriate output measure. In both industries, it is important to know how many labor hours are required to produce one unit of output. This measure can be used to compare the labor efficiencies of different companies in a given industry, or to compare the same industries among different nations. Obviously, fewer labor hours are better and mean higher productivity. A company or a country that can produce the same output with fewer input labor hours not only has a higher productivity; it also has a competitive advantage in the global economy.

Although labor productivity is a commonly used measure, labor itself does not contribute much to improving productivity. More important factors in determining and improving productivity are capital and technology. ***Capital*** refers to the substitution of machines for human labor; for example, investing in an automated production machine to replace a manually operated machine. The automated machine can probably operate at a higher production rate, so even if a worker is still needed to monitor the operation, productivity has been increased. If the worker is no longer needed at the machine, then labor productivity has been increased even more. ***Technology*** refers to a fundamental change in the way some activity or function is accomplished. It is more than simply using a machine in place of a human worker. It is using a brand new type of machine to replace the previous type. Some examples of how new technologies have made dramatic improvements in productivity are listed in Table 7.

Admittedly, the distinctions between capital improvements and technology improvements are sometimes subtle, because new technologies almost always require capital investments. Anyway, arguing about these differences and subtleties is not as important as recognizing that the important gains in productivity are generally made by the introduction of capital and technology in a work process, rather than by attempting to extract more work in less time from workers. By investing in capital and technology to increase the rate of output work units and/or reduce input labor hours, the labor productivity ratio is increased.

TABLE 7 Examples of Technology Changes that Dramatically Improved Productivity

Old Technology	New Technology	Improvement
Horse-drawn carts	Railroad trains	Substitution of steam power for horse power, use of multiple carts (passenger or freight cars)
Steam locomotive	Diesel locomotive	Substitution of diesel power technology for steam power technology
Telephone operator	Dial phone	Dial technology allowed "clicks" of the dial to be used to operate telephone switching systems
Dial phone	Touch-tone phone	Substitution of tone frequencies in place of "clicks" to operate telephone switching systems for faster dialing
Manually operated milling machine	Numerical control milling machine	Substitution of coded numerical instructions to operate the milling machine rather than a skilled machinist
DC3 passenger airplane (1930s)	Boeing 747 passenger airplane (1980s)	Substitution of jet propulsion for piston engine for higher speed, larger aircraft for more passenger miles

Measuring productivity is not as easy as it seems. Although the labor productivity ratio defined by equation (1) appears simple and straightforward, the following problems are often encountered in its measurement and use:

- *Nonhomogeneous output units.* The output work units are not necessarily homogeneous. For example, using the annual production of automobiles as the output measure does not account for differences in models, vehicle sizes, and prices. An expensive luxury model is likely to require more labor hours of assembly time than an inexpensive compact car.
- *Multiple input factors.* As previously indicated, labor is not the only input factor in determining productivity. In addition to capital and technology, other input factors may include materials and energy. For example, in the production of aluminum, electric power and the raw material bauxite are much more important than labor as inputs to the process.[6]
- *Price and cost changes.* The prices of output work units and the costs of input factors (labor, materials, power) change over time, often unpredictably. A company may improve productivity, but if the prices of its products decrease due to market forces, the company could find itself in severe financial difficulty. The steel industry in the United States during the late 1990s and early 2000s provides a perfect example of this case.
- *Product mix changes.* Product mix refers to the relative proportions of products that a company sells. If the mix of expensive and inexpensive products changes from year to year, an annual comparison based on the labor productivity ratio is less meaningful.

An alternative productivity measure is the labor productivity index that compares the output/input ratio from one year to the next. The productivity index is defined as follows:

$$LPI = \frac{LPR_t}{LPR_b} \tag{2}$$

where LPI = labor productivity index, LPR_t = labor productivity ratio during some time period of interest, and LPR_b = labor productivity ratio during some defined base period.

Example 1 Productivity Measurement

During the base year in a small steel mill, 326,000 tons of steel were produced using 203,000 labor hours. In the next year, the output was 341,000 tons using 246,000 labor hours. Determine (a) the labor productivity ratio for the base year, (b) the labor productivity ratio for the second year, and (c) the labor productivity index for the second year.

Solution **(a)** In the base year, $LPR = \dfrac{326{,}000}{203{,}000} = 1.606$ tons per labor hour.

(b) In the second year, $LPR = \dfrac{341{,}000}{246{,}000} = 1.386$ tons per labor hour.

[6]Bauxite is the principal ore used to produce aluminum. It consists largely of hydrated aluminum oxide ($Al_2O_3 H_2O$).

(c) The productivity index for the second year is $LPI = \dfrac{1.386}{1.606} = 0.863$.

Comment: No matter how it is measured, productivity went down in the second year. ■

4.2 Productive Work Content

Work is usually not performed in the most efficient way possible. A given task performed by a worker can be considered to consist of (1) the basic productive work content and (2) excess nonproductive activities. In general, the task is associated with the production of a product or the delivery of a service. The ***basic productive work content*** is the theoretical minimum amount of work required to accomplish the task, where the amount of work is expressed in terms of time. Thus, the time required to perform the basic productive work content is the theoretical minimum. It cannot be further reduced. The basic productive work content time is a theoretical concept implying optimal conditions that rarely, if ever, are achieved in practice. Nevertheless, it is an optimum worth seeking.

The ***excess nonproductive activities*** in the task are the extra physical and mental actions performed by the worker that do not add any value to the task, nor do they facilitate the productive work content that does add value. The excess nonproductive activities take time. They add to the basic productive work content time to make up the total time needed to perform the task. The excess nonproductive activities can be classified into three categories, as illustrated in the time line in Figure 4:

- Excess activities caused by poor design of the product or service
- Excess activities caused by inefficient methods, poor work layout, and interruptions
- Excess activities caused by the human factor

Some examples of the three categories of nonproductive activities are listed in Table 8. As several of these examples indicate, nonproductive activities do not necessarily occur during every cycle of the task. For example, industrial accidents occur infrequently in most work situations. However, over the long run, their effects extract a heavy toll on productive work content.

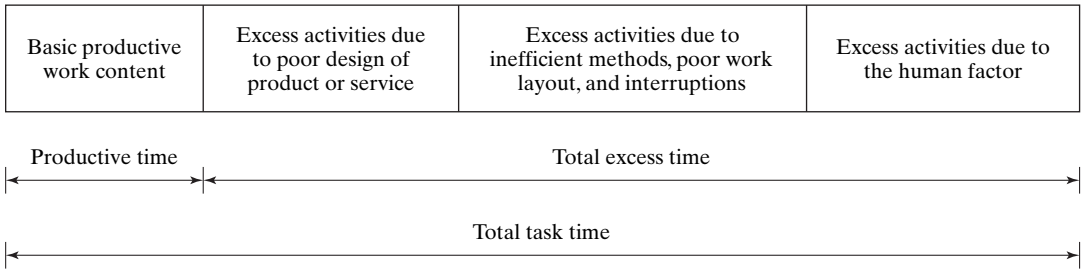


Figure 4 Total task time consists of basic productive work content and nonproductive activities.

TABLE 8 Examples of Excess Nonproductive Activities that Add to Total Task Time

Category of Activity	Examples
Poor design of the product or service	Products designed with more parts than necessary, so that excess assembly time is required Product proliferation (e.g., more choices for the customer than necessary) Frequent product design changes, causing changes in tooling, methods, and layout Waste of materials (e.g., machining from bar stock rather than machining from a forging or casting) Quality standards that are more stringent than necessary, requiring excess processing time
Inefficient methods, poor work layout, and interruptions	Inefficient plant layout that requires excess movement of materials Inefficient workplace layout that requires excess hand, arm, and body motions Inefficient methods that waste time Wasted space (e.g., using valuable floor space to store materials instead of storing vertically in racks) Inefficient material handling (e.g., using manual methods to move materials rather than mechanized methods) Long setup times between batches of work Frequent breakdowns of equipment Workers waiting for work
The human factor	Absenteeism and tardiness Workers spending too much time socializing Workers deliberately working slowly Inadequate training of workers Industrial accidents Hazardous materials that cause occupational illnesses

Source: Adapted from [9].

The concepts of basic productive work content and excess nonproductive activities are applicable not only to individual tasks but also to the entire work sequence required to manufacture a product or provide a service. There are value-adding activities and non-value-adding activities in virtually all work systems. An important objective in the design of a work system is to minimize the non-value-adding activities so that only productive work is accomplished. This objective is achieved through the use of methods engineering, operations analysis, work measurement, and other topics discussed in this text.

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REVIEW QUESTIONS

1 Define work.

2 What are basic motion elements? Give some examples.

3 What is a work element?

4 Why is time important in work?

5 Define work system as a physical entity.

6 Define work system as a field of professional practice.

7 What are some of the functions included within the scope of work management?

8 Name the four broad categories of worker occupations.

9 Define productivity.

10 Labor is one input factor that determines productivity. What are two other factors that are more important than labor in improving productivity? Define each of these two additional input factors.

11 What is the difference between the labor productivity ratio and the labor productivity index?

12 A given task performed by a worker can be considered to consist of the basic productive work content and excess nonproductive activities. (a) What is meant by the term *basic productive work content*? (b) What is meant by the term *excess nonproductive activities*?

13 What are the three categories of excess nonproductive activities, as they are defined in the text?

PROBLEMS

1 A work group of 5 workers in a certain month produced 500 units of output working 8 hr/day for 22 days in the month. (a) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? (b) Suppose that in the next month, the same work group produced 600 units but there were only 20 workdays in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.

2 A work group of 10 workers in a certain month produced 7200 units of output working 8 hr/day for 22 days in the month. Determine the labor productivity ratio using (a) units of output per worker-hour and (b) units of output per worker-month. (c) Suppose that in the next month, the same work group produced 6800 units but there were only 20 workdays in the month. For each productivity measure as in (a) and (b), determine the productivity index for the next month using the prior month as a base.

3 A work group of 20 workers in a certain month produced 8600 units of output working 8 hr/day for 21 days. (a) What is the labor productivity ratio for this month? (b) In the next month, the same work group produced 8000 units but there were 22 workdays in the month and the size of the work group was reduced to 14 workers. What is the labor productivity ratio for this second month? (c) What is the productivity index using the first month as a base?

4 There are 20 forging presses in the forge shop of a small company. The shop produces batches of forgings requiring a setup time of 3.0 hours for each production batch. Average standard time for each part in a batch is 45 seconds, and there are an average of 600 parts in a batch. The plant workforce consists of two workers per press, two foremen, plus three clerical support staff. (a) Determine how many forged parts can be produced in 1 month, if there are 8 hours worked per day and an average of 21 days per month at one shift per day. (b) What is the labor productivity ratio of the forge shop, expressed as parts per worker-hour?

5 A farmer's market is considering the addition of bar code scanners at their check-out counters, which would use the UPC marked on all grocery packaging. Currently, the check-out clerk keypunches the price of each item into the register during check-out. Observations indicate that an average of 50 items are checked out per customer. The clerk currently takes 7 seconds per item to keypunch the register and move the item along the check-out table. On average it takes 25 seconds to total the bill, accept money from the customer, and make change. It then takes 4 seconds per item for the clerk to bag the customer's order. Finally, about 5 seconds are lost to transition to the next customer. Bar code scanners would eliminate the need to keypunch each price, and the time per item would be reduced to 3 seconds with the bar code scanner. (a) What is the hourly throughput rate (number of customers checked out per hour) under the current check-out procedure? (b) What would be the estimated hourly throughput rate if bar code scanners were used? (c) If separate baggers were used instead of requiring the check-out clerk to perform bagging in addition to check-out, what would be hourly throughput rate? Assume that bar code scanners are used by the clerk. (d) Determine the productivity index for each of the two cases in (b) and (c), using (a) as the basis of comparison and hourly customers checked out per labor hour as the measure of productivity.

Manual Work and Worker–Machine Systems

1 Manual Work Systems
  1.1 Types of Manual Work
  1.2 Cycle Time Analysis of Manual Work
2 Worker–Machine Systems
  2.1 Types of Worker–Machine Systems
  2.2 Cycle Time Analysis in Worker–Machine Systems
3 Automated Work Systems
4 Determining Worker and Machine Requirements
  4.1 When Setup Is Not a Factor
  4.2 When Setup Is Included
5 Machine Clusters

There are various types of work systems used in production, services, offices, projects, and other work situations. All of these work systems utilize the physical and mental capabilities of humans. In terms of the human participation in the tasks performed, work systems can be classified into the three basic categories depicted in Figure 1: (a) manual work systems, (b) worker–machine systems, and (c) automated systems. A ***manual work system*** consists of a worker performing one or more tasks without the aid of powered tools. The tasks commonly require the use of hand tools (e.g., hammers, screwdrivers, shovels). In a ***worker–machine system***, a human worker operates powered equipment (e.g., a machine tool). An ***automated work system*** is one in which a process is performed by a machine without the direct participation of a human worker.

As indicated in Figure 1, the work accomplished by a work system is almost always acted upon some object, called the ***work unit***. The state of the work unit is advanced in some way through the process performed on it. In production, the work may alter the geometry of a work part. In logistics, the work may involve transporting

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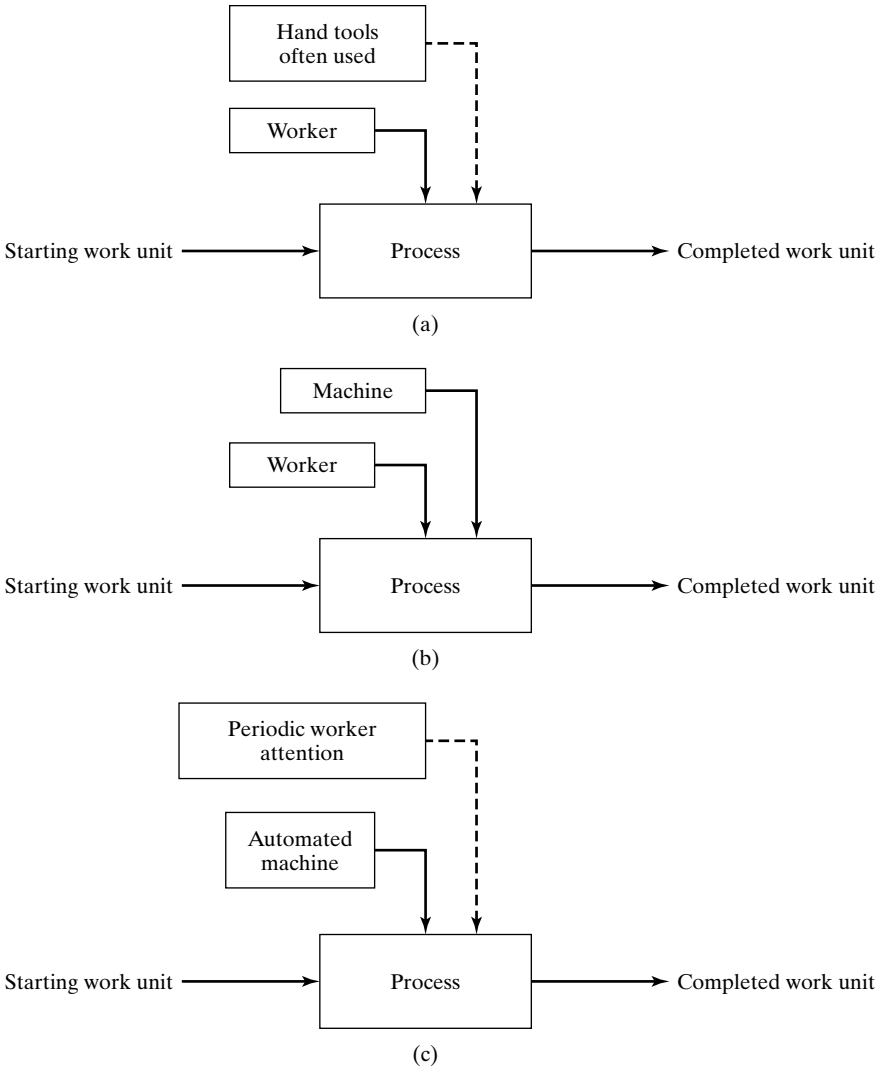


Figure 1 Three types of work systems: (a) manual work system, (b) worker–machine system, and (c) automated system.

material from a warehouse to a customer. In service work, a sales prospect is transformed into a paying customer by a persuasive salesperson. In knowledge work, a designer takes a product concept and converts it into specifications and engineering drawings.

In this chapter, we discuss the three categories of work systems. These systems are associated mostly with work that is performed by production and logistics workers. The emphasis in our coverage is on unit operations—tasks and

processes that are treated as being independent of other work activities in a given facility or work site.

1 MANUAL WORK SYSTEMS

Manual work is the most basic form of work, engaging the human body to accomplish some physical task without an external source of power. Hand tools are often used to facilitate the task, but the power to operate them derives from the strength and stamina of a human worker. With or without hand tools, the worker must expend physical energy to accomplish the task. In addition, other human faculties are required, such as hand-eye coordination and mental effort. Our coverage of manual work includes two sections: (1) types of manual work and (2) cycle time analysis of manual work.

1.1 Types of Manual Work

Two forms of manual work can be distinguished: (1) pure manual work and (2) manual work using hand tools. ***Pure manual work*** involves only the physical and mental capabilities of the human worker, and no machines, tools, or other implements are employed in performing the task. Examples of pure manual work include the following:

- A material-handling worker moving cartons from the floor onto a conveyor in a warehouse
- Workers loading furniture into a moving van from a house without the use of dollies or other wheeled platforms
- A dealer at a casino table dealing cards
- An office worker filing documents in a file cabinet
- An assembly worker snap-fitting two parts together
- An assembly worker assembling two sheet metal parts with a bolt and nut by hand (tightening is done later using appropriate hand or power tools).

Note that the common characteristic in these examples is that they consist of moving things. Performing manual work without tools almost always involves the movement and handling of objects. Even assembly tasks include moving the parts in order to join them.

Manual tasks are commonly augmented by the use of hand tools. The ability to design and use tools is one of the attributes that distinguish humans from other species on earth. A ***tool*** is a device or implement for making changes to some object (e.g., the work unit), such as cutting, grinding, striking, squeezing, or other process. Instruments used for measurement are also included in the category of tools, even though no physical change in the object results directly from their use. A ***hand tool*** is a small tool that is operated by the strength and skill of the human user. When using hand tools, a ***workholder*** is sometimes employed to grasp the work unit and position it securely during

processing. Examples of manual tasks involving the use of hand tools include the following:

- A machinist using a file to round the edges of a rectangular part that has just been milled
- An assembly worker using a screwdriver to tighten a screw
- A painter using a paintbrush to paint the trim around a doorway
- A sculptor using a carving knife to carve a wooden statue
- A grass-cutter using a rake to collect the grass clippings after mowing
- A quality control inspector using a micrometer to measure the diameter of a shaft
- A material-handling worker using a dolly to move cartons in a warehouse
- An office worker using a pen to handwrite entries into a ledger.

1.2 Cycle Time Analysis of Manual Work

Manual tasks usually consist of a work cycle that is repeated with some degree of similarity, and each cycle usually corresponds to the processing of one work unit. When a painter is hired to paint the wooden trim around the doorways in a new house, the painting cycle is repeated for each doorway. If the doorways are all the same size, and the wood trim is identical for all doors, then the painting cycle should exhibit a high degree of similarity. If there are differences in the doorways, then the painting cycles will be less similar.

If the work cycle is relatively short, and there is a high degree of similarity from one cycle to the next, we refer to the work as ***repetitive***. If the work cycle takes a long time and the cycles are not similar, the work is ***nonrepetitive***.[1] In either case, the task can be divided into work elements that consist of logical groupings of motions performed by the worker. The cycle time T_c is therefore the sum of the work element times

$$T_c = \sum_{k=1}^{n_e} T_{ek} \tag{1}$$

where T_{ek} = time of work element k, where k is used to identify the work elements, min; and n_e = number of work elements into which the cycle is divided. Our focus in this section is on repetitive work. Consider the following example of a pure manual task.

Example 1 A Repetitive Manual Task

An assembly worker performs a repetitive manual task consisting of inserting 8 plastic pegs into 8 holes in a flat wooden board. A slight interference fit is involved in each insertion. The worker holds the board in one hand and picks up the pegs from a tray with the other hand and inserts them into the holes, one peg at a time. The workplace layout is shown in Figure 2 (a), and the sequence of work elements is given in the table below. Can the work method be improved in order to reduce the cycle time?

[1]There is no clear boundary between repetitive and nonrepetitive work. When referring to repetitive work cycles, we usually mean a cycle time of a few minutes or less, and the motion patterns are intended to be identical for every cycle, so that cycle-to-cycle variations in time and work content tend to be random.

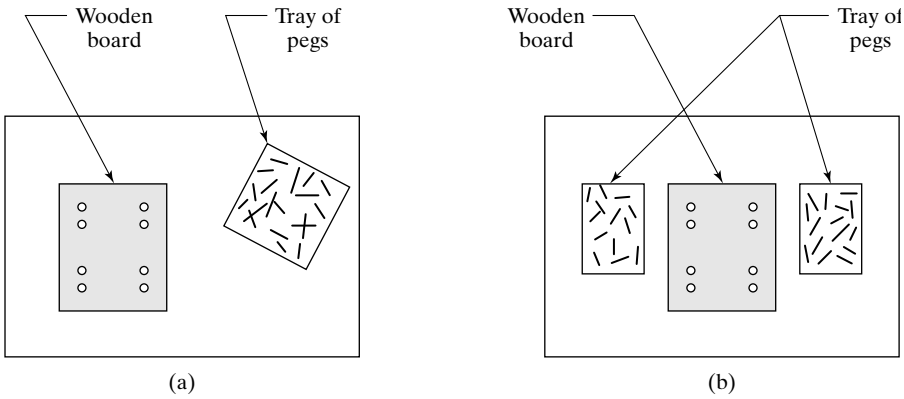


Figure 2 Workplace layout for assembly task in Example 1: (a) before methods improvement and (b) after methods improvement.

Sequence	Work Element Description	Work Element Time, T_{ek} (min)
1	Worker picks up board with one hand and holds it.	0.08
2	Worker picks peg from tray and inserts it into hole in board.	0.06
3	Worker picks second peg and inserts it into hole in board.	0.06
4	Worker picks third peg and inserts it into hole in board.	0.06
5	Worker picks fourth peg and inserts it into hole in board.	0.06
6	Worker picks fifth peg and inserts it into hole in board.	0.06
7	Worker picks sixth peg and inserts it into hole in board.	0.06
8	Worker picks seventh peg and inserts it into hole in board.	0.06
9	Worker picks eighth peg and inserts it into hole in board.	0.06
10	Worker lays assembled board into tote pan.	0.06
Total work cycle time		0.62

Solution: An opportunity for improvement lies in using a work-holding device to hold and position the board while the worker uses both hands simultaneously to insert the pegs. Two trays filled with pegs will be used, one for each hand. Instead of picking the pegs out of the tray one peg at a time, each hand will grab four pegs in order to minimize the number of times the worker's hands must reach to the trays. The revised workplace layout is shown in Figure 2 (b), and the revised sequence of work elements is presented in the table below. The cycle time is reduced from 0.62 min to 0.37 min, a reduction of 40%, which corresponds to an increase in production rate of almost 68%.

Sequence	Work Element Description	Work Element Time, T_{ek} (min)
1	Worker picks up board and positions it in workholder.	0.12
2	Worker picks 4 pegs each with both hands from 2 trays and inserts them into 8 holes in board.	0.15
3	Worker removes board from workholder and places in tote pan.	0.10
Total work cycle time		0.37

■

As the example illustrates, it is important to design the work cycle so as to minimize the time required to perform it. There are many alternative ways to perform a given task, where the differences are in work elements, hand and body motions, tools, workholders, and so forth. Some of the alternative methods are less time-consuming than others. A useful concept in work design is the ***one best method*** principle. According to this principle, of all the possible methods that can be used to perform a task, there is one optimal method that minimizes the time and effort required to accomplish it.[2] One of the primary objectives in work design is to determine the one best method for the task, and then to standardize its use in the workplace.

Normal Time and Standard Time. Once the work cycle and associated method are defined, the actual time taken for a given manual cycle is a variable. Variability is inherent in any repetitive human activity, and this variability is manifested in the time to perform the activity. Reasons for variations in work cycle times include the following:

- Differences in worker performance from one cycle to the next
- Variations in hand and body motions
- Worker blunders and bungles
- Variations in the starting work units
- Inclusion of extra elements that are not performed every cycle
- Differences in the physical and cognitive attributes among workers performing the task
- Variations in the methods used by different workers to perform the task
- The learning curve phenomenon.

Topping the list is worker ***performance***, which can be defined simply as the pace or relative speed with which the worker does the task. As worker performance increases, the time to accomplish the work cycle decreases. From the employer's viewpoint, it is desirable for the worker to work at a high level of performance. The question is: what is a reasonable performance or pace to expect from a worker in accomplishing a given task? To answer the question, let us introduce the concept of normal performance. ***Normal performance*** (or ***normal pace***) means a pace of working that can be maintained by a properly trained average worker throughout an entire work shift without deleterious short-term or long-term effects on the worker's health or physical well-being. The work shift is usually assumed to be eight hours, during which periodic rest breaks are allowed. Normal performance refers to the pace of the worker while actually working. When a worker works at a normal performance level, we say he or she is working at 100% performance. A faster pace than normal is greater than 100% and a slower pace is less than 100%. A common benchmark of normal performance is walking on level ground at three miles per hour.

The term ***standard performance*** is often used in place of normal performance. They both refer to the same pace while working, but standard performance acknowledges that periodic rest breaks are included in the work shift. For example, a healthy human

[2]The one best method should also satisfy other criteria in addition to minimizing the work cycle time, such as ensuring a safe and convenient workplace for the worker, and producing a work unit of high quality.

can walk at a pace of three miles per hour for an hour or two. However, walking for a solid eight hours without ever stopping for a rest break would be physically wearing. Accordingly, normal performance and standard performance both mean walking three miles per hour during an eight-hour period, but the walker could stop for a rest a few times and take a lunch break during the eight hours.

When a work cycle is performed at 100% performance, the time taken is called the ***normal time*** for the cycle. If worker performance is greater than 100%, then the time required to complete the cycle will be less than the normal time; and when worker performance is less than 100 percent, the time taken will be greater than the normal time. The actual time to perform the work cycle is a function of the worker's performance as indicated in the equation

$$T_c = \frac{T_n}{P_w} \tag{2}$$

where T_c = actual cycle time, min; T_n = normal time for the work cycle, min; and P_w = pace or performance of the worker, expressed as a decimal fraction (e.g., 100% = 1.0).

Example 2 Normal Performance

A man walks in the early morning for health and fitness. His usual route is 1.85 miles long. The route has minimal elevation changes. A typical time to walk the 1.85 miles is 30 min. Using the benchmark of 3 miles/hr as normal performance, determine (a) how long the route would take at normal performance and (b) the man's performance when he completes the route in 30 min.

Solution: **(a)** At 3 miles/hr, 1.85 miles can be covered in 1.85/3.0 = 0.6167 hr or 37 min.

(b) If the man takes 30 min to complete the walk, then his performance can be determined by dividing the normal time by the actual time, by rearranging equation (2). Thus,

$$P_w = 37/30 = 1.233 \text{ or } 123.3\%$$

Alternative Solution: For part (b), we could determine the man's velocity and compare it to the benchmark of 3.0 miles/hr in order to determine his performance. If the man completes 1.85 miles in 30 min, then his walking velocity is 1.85 miles divided by 0.5 hr, which equals 3.7 miles/hr.

$$P_w = 3.7/3.0 = 1.233 \text{ or } 123.3\%$$ ■

Workers are allowed periodic rest breaks (e.g., coffee breaks) during their work shift. A typical work shift is eight hours (e.g., 8:00 A.M. to 5:00 P.M. with an hour from noon to 1:00 P.M. for lunch). The shift usually includes a rest break in the morning and another in the afternoon. Unlike the lunch period, these rest breaks are normally included within the eight-hour time of the shift. The employer allows these breaks because it has been found that the overall productivity of the worker during the shift is greater if rest breaks are provided. More work is accomplished by the end of the day and fewer mistakes are made if the worker can take time out periodically from the normal work routine. In addition to the rest breaks, the worker is likely to have other interruptions during the

shift, such as equipment breakdowns (if the manual task is somehow dependent on equipment), receiving instructions from the foreman, personal telephone calls, and so on. As a result of all of these factors, the total time actually worked during the shift will be less than the full eight hours, in all likelihood.

To account for these delays and rest breaks, an ***allowance*** is added to the normal time in order to determine an "allowed time" for the worker to perform the task throughout the shift. More commonly known as the ***standard time***, it is defined as follows:

$$T_{std} = T_n(1 + A_{pfd}) \tag{3}$$

where T_{std} = standard time, min; T_n = normal time, min; and A_{pfd} = allowance factor, usually expressed as a percentage but used in equation (3) as a decimal fraction. The allowance is commonly called the personal time, fatigue, and delay allowance (abbreviated ***PFD allowance***), and it is figured in such a way that, if the worker works at 100 percent performance during the portion of the shift that he or she is working, the amount of work accomplished will be eight hours' worth.

Manual work cycles often include ***irregular work elements***, which are elements performed with a frequency of less than once per cycle. Examples of irregular work elements include periodic changing of tools (e.g., changing a knife blade) and replacing tote pans of parts when the containers become full. In determining a standard time for the cycle, the irregular element times are prorated in the regular cycle time. The following examples illustrate these concepts and definitions.

Example 3 Determining Standard Time and Standard Output

The normal time to perform the regular work cycle for a certain manual operation is 3.23 min. In addition, an irregular work element whose normal time is 1.25 min must be performed every 5 cycles. The PFD allowance factor is 15%. Determine (a) the standard time and (b) how many work units are produced if the worker's performance in an 8-hour shift is standard.

Solution: **(a)** The normal time for the work cycle includes the irregular element prorated according to its frequency:

$$T_n = 3.23 + \frac{1.25}{5} = 3.23 + 0.25 = 3.48 \text{ min}$$

The standard time is $T_{std} = T_n(1 + A_{pfd}) = 3.48(1 + 0.15) = 4.00$ min.

(b) The number of work units produced at standard performance in an 8-hour shift is the clock time of the shift divided by the standard time:

$$Q_{std} = \frac{8.0(60)}{4.00} = 120 \text{ work units}$$

■

Example 4 Determining Lost Time Due to the Allowance Factor

Determine the anticipated amount of time lost per 8-hour shift when an allowance factor of 15% is used, as in the previous example.

Solution: Given that $A_{pfd} = 0.15$, the anticipated amount of time lost per 8-hour shift is determined as follows:

$$8.0 \text{ hr} = (\text{actual time worked})(1 + 0.15)$$

$$\text{Actual time worked} = \frac{8.0}{1.15} = 6.956 \text{ hr}$$

$$\text{Time lost} = 8.0 - 6.956 = 1.044 \text{ hr}$$

This is the anticipated daily amount of time lost due to personal time, fatigue, and delays corresponding to a 15% PFD allowance factor. ■

Example 5 Production Rate When Worker Performance Exceeds 100%

Now that the standard is set ($T_{std} = 4.00$ min), and given the data from the previous examples, how many work units would be produced if the worker's average performance during an 8-hour shift were 125% and the hours actually worked were exactly 6.956 hr, which corresponds to the 15% allowance factor.

Solution: Based on the normal time $T_n = 3.48$ min, the actual cycle time with a worker performance of 125% is

$$T_c = \frac{3.48}{1.25} = 2.78 \text{ min}$$

Assuming one work unit is produced each cycle, the corresponding daily production rate (symbolized by R_p) is

$$R_p = \frac{6.956(60)}{2.78} = 150 \text{ work units}$$

Note that 150 units = 125% of 120 units at 100% performance. ■

Standard Hours and Worker Efficiency. Two common measures used in industry to assess a worker's productivity are standard hours and worker efficiency. The ***standard hours*** represents the amount of work actually accomplished by the worker during a given period (e.g., shift, week), expressed in terms of time. In its simplest form, it is the quantity of work units produced during the period multiplied by the standard time per work unit; that is,

$$H_{std} = QT_{std} \tag{4}$$

where H_{std} = standard hours accomplished, hr; Q = quantity of work units completed during the period, pc; and T_{std} = standard time per work unit, hr/pc. If the time standard

T_{std} is expressed in min/pc, then conversion of units is required to obtain standard hours H_{std}. When a worker works at a performance level greater than 100% and his or her actual time worked during the shift is consistent with or greater than what is provided by the allowance factor, then the number of standard hours accomplished will be greater than the number of hours in the shift.

Worker efficiency is the amount of work accomplished during the shift expressed as a proportion of the shift hours. In equation form,

$$E_w = \frac{H_{std}}{H_{sh}} \tag{5}$$

where E_w = worker efficiency, normally expressed as a percentage; H_{std} = number of standard hours of work accomplished during the shift, hr; and H_{sh} = number of shift hours (e.g., 8 hr).

Example 6 Standard Hours and Worker Efficiency

For the worker performance of 125% in the previous example (T_{std} = 4.00 min), determine (a) number of standard hours produced and (b) worker efficiency.

Solution: **(a)** $H_{std} = 150(4.0 \text{ min}) = 600 \text{ min} = 10.0 \text{ hr}$
(b) $E_w = (10.0 \text{ hr})/(8.0 \text{ hr}) = 1.25 = 125\%$ ■

In Example 5 and 6, the worker's efficiency and performance level are equal for two reasons: (1) the number of hours actually worked is exactly consistent with the 15% allowance factor, and (2) the entire work cycle consists exclusively of manual labor and is therefore entirely operator-controlled. In the absence of either or both of these conditions, worker efficiency will not equal worker performance level (except by coincidence, when certain combinations of values offset each other). In reality, the number of hours actually worked by a worker in an 8-hour shift varies each day, depending on the amount of time lost due to personal reasons, rest breaks, and delays. Worker performance and worker efficiency are different if the time lost is different from what is accounted for by the PFD allowance factor, as the following example illustrates.

Example 7 Standard Hours and Worker Efficiency as Affected by Hours Actually Worked

Suppose the worker's pace in the task is 125%, but the actual hours worked is 7.42 hr. Determine (a) the number of pieces produced, (b) the number of standard hours accomplished, and (c) the worker's efficiency.

Solution: **(a)** The actual cycle time at 125% performance is 2.78 min, as calculated in Example 5. The number of work units produced in 7.42 hr is

$$Q = \frac{7.42(60)}{2.78} = 160 \text{ units}$$

(b) $H_{std} = 160(4.0 \text{ min}) = 640 \text{ min} = 10.67 \text{ hr}$
(c) $E_w = 10.67/8.0 = 1.333 = 133.3\%$ ■

Worker efficiency is commonly used to evaluate workers in industry. In many incentive wage payment plans, the worker's earnings are based on his or her efficiency or the

number of standard hours accomplished. Worker efficiency and standard hours are easily computed, because the number of hours in the shift and the standard time are known, and the number of work units produced can be readily counted. The two measures are basically equivalent, because either one can be derived from the other. One might think that worker performance would also be a useful measure; however, it is more difficult to assess because it requires data on the amount of time actually worked by the worker during the shift. This varies from day to day because the interruptions and delays vary from day to day. Some method of continuously observing the worker would be required. Aside from the cost, the worker would likely find such observation objectionable. In addition, the effect of worker performance is reduced when machine time is included in the work cycle, as we see in Section 2.

2 WORKER–MACHINE SYSTEMS

When a worker operates powered equipment, we refer to the arrangement as a ***worker–machine system***. It is one of the most widely used work systems. Worker–machine systems include combinations of one or more workers and one or more pieces of equipment. The workers and machines are combined to accomplish a desired output. Examples of worker–machine systems include the following:

- A skilled machinist operating an engine lathe in a tool room to fabricate a component (the work unit) for a custom-designed product. The machinist must exercise considerable skill in controlling the feed, speed, and tool position while operating the lathe.
- A construction worker operating a backhoe at a construction site. The worker must continuously operate the machine, using the various levers that control the different hydraulically operated mechanisms.
- A truck driver driving an 18-wheel tractor-trailer on an interstate highway. The driver must constantly be alert while operating the vehicle.
- A factory worker loading and unloading parts at a machine tool. The machine tool operates on semiautomatic cycle to process the parts. At the end of each work cycle, the worker unloads the completed part. Machine processing time is about 3 min. While the machine performs its process, the worker is idle. Loading and unloading the machine takes about 30 sec.
- A crew of workers operating a rolling mill that converts hot steel slabs into flat plates. Each worker has an assigned function. The most important job is the rolling mill operator who must coordinate the gap size (i.e., distance between opposing rolls) and the passing of the slab back and forth between the rolls. Each pass reduces the thickness of the starting slab until the specified thickness has been achieved.
- A secretary using a personal computer with word processor in an office typing pool.
- A clerical worker in a billing center entering data based on checks received by mail from customers into account records on a networked personal computer.
- An industrial engineer creating the design of a plant layout on a computer-aided design (CAD) workstation.

TABLE 1 Relative Strengths and Attributes of Humans and Machines

Relative Strengths of Humans	Relative Strengths of Machines
Sense unexpected stimuli	Perform repetitive tasks consistently
Develop new solutions to problems	Store large amounts of data
Cope with abstract problems	Retrieve data from memory reliably
Adapt to change	Perform multiple tasks at the same time
Generalize from observations	Apply high forces and power
Learn from experience	Perform simple computations quickly
Make difficult decisions based on incomplete data	Make routine decisions quickly

Source: [2].

Although the last three examples relate to service and knowledge work rather than production and logistics work, they also illustrate the widespread use of worker–machine systems. In these latter examples, the machine is a computer.

In a worker–machine system, the worker and the machine both contribute their own strengths and capabilities to the combination, and the result is synergistic. The relative strengths and attributes of humans and machines are presented in Table 1, and the worker–machine system should be designed to exploit these relative strengths.

2.1 Types of Worker–Machine Systems

It is instructive to distinguish the various categories and arrangements of worker–machine systems, some of which are suggested by our list of examples. In this section, we discuss the following classifications: (1) types of powered machinery used in the system, (2) numbers of workers and machines in the system, and (3) level of operator attention required to run the machinery.

Types of Powered Machinery. Powered machinery is distinguished from hand tools by the fact that a source of power other than human (or animal) strength is used to operate it. Common power sources are electric, pneumatic, hydraulic, and fossil fuel motors (e.g., gasoline, propane). In most cases, the power source is converted to mechanical energy to process the work unit. Powered machinery can be classified into three categories, summarized in Figure 3: (1) portable power tools, (2) mobile powered equipment, and (3) stationary powered machines.

Portable power tools are light enough in weight that they can be carried by the worker from one location to another and manipulated by hand. Examples include portable power drills, rotary saws, chain saws, and electric hedge trimmers. Common power sources are electric, pneumatic, and gasoline.

Mobile powered equipment can be divided into three categories: (1) transportation equipment, (2) transportable and mobile during operation, and (3) transportable and stationary during operation. They are generally heavy pieces of equipment and cannot be classified as power hand tools. Transportation equipment is a large category that includes cars, taxicabs, buses, trucks, trains, airplanes, boats, and ships. This powered machinery is designed to carry materials and/or people. Equipment in category (2), transportable

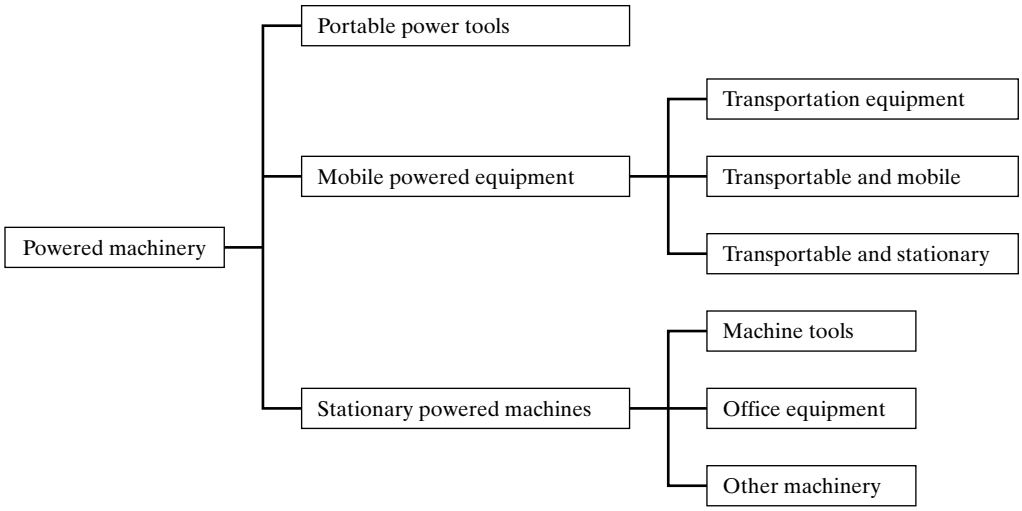


Figure 3 Classification of powered machinery in worker–machine systems.

and mobile during operation, consists of equipment that can move under its own power but can also be moved by transportation equipment (e.g., tractor and flatbed). Examples include construction equipment (e.g., bulldozers, backhoes), agricultural and lawn-keeping (e.g., small tractors, lawn mowers), and material-handling equipment (e.g., forklift trucks). The third category, transportable and stationary during operation, is equipment that can be transported by highway truck but it performs its function in a stationary location once it is moved (e.g., electric power generator, large power saws used at construction sites). The typical power sources are fossil fuels.

Stationary powered machines stand on the floor or ground and cannot be moved while they are operating, and they are not normally moved between operations. Electricity is the usual power source. We can classify stationary power tools into the following categories: (1) machine tools, (2) office equipment, and (3) other. A ***machine tool*** is a stationary power-driven machine that shapes or forms parts. Machine tools are normally associated with factory production operations such as machining (e.g., turning, drilling, milling), shearing (e.g., blanking, hole-punching), and squeezing (e.g., forging, extrusion). Office equipment (second category) includes personal computers, photocopiers, telephones, fax machines, design workstations, and other equipment and systems normally found in an office facility. The "other" types of equipment (third category) are a miscellaneous group that includes machinery not fitting into the other two categories. Examples include furnaces, ovens, cash registers, and sewing machines.

Numbers of Workers and Machines. Another means of classifying worker–machine systems is according to whether there are one or more workers and one or more machines. This provides four categories, as indicated in Table 2.

TABLE 2 Classification of Worker–Machine Systems According to Number of Machines and Workers

	One Machine	Multiple Machines
One worker	One worker–one machine. **Examples**: (1) A worker loading and unloading a machine tool, (2) a truck driver driving a tractor trailer.	One worker–multiple machines **Example**: A worker tending several production machines.
Multiple workers	Multiple workers–one machine. **Examples**: (1) Several workers operating a rolling mill, (2) a crew on a ship or airplane.	Multiple worker–multiple machines. **Example**: An emergency repair crew responding to machine breakdowns in a factory.

For the case of one worker and one machine, good work design attempts to achieve the following objectives:

- Design the controls of the machine to be logical and easy to operate for the worker.
- Design the work sequence so that as much of the worker's task as possible can be accomplished while the machine is operating, thereby minimizing worker idle time.
- Minimize the idle times of both the worker and the machine.
- Design the task and the machine to be safe for the worker. If the task is inherently hazardous to the worker, then an automated work system should be considered.

The same design objectives are applicable when the work system consists of multiple workers and/or multiple machines. An additional objective is to optimize the number of workers or machines in the system according to some appropriate economic objective. For example, if one worker is assigned to attend to multiple machines, how many machines should be assigned to that worker so as to avoid machine idle time?

Level of Operator Attention Required. Another way to distinguish among worker–machine systems is by the level of attention required by the worker(s). In this classification, we have the four categories described in Table 3. Full-time attention means that the worker must devote virtually 100% of his or her time to the operation of the equipment during the performance of the task.

2.2 Cycle Time Analysis in Worker–Machine Systems

In terms of cycle time analysis, worker–machine systems fall into two categories: (1) systems in which the machine time depends on operator control, and (2) systems in which the machine time is constant and independent of operator control, and the work cycle is repetitive. In the first category, in which machine time depends on operator control, the task can be either (1) repetitive or (2) nonrepetitive. The following examples illustrate repetitive tasks with cycle times that depend on the pace and skill with which the operator applies the powered equipment:

- A typist typing a list of names and telephone numbers on a conventional electric typewriter

TABLE 3 Levels of Operator Attention in Worker–Machine Systems

Category	Description	Examples
Full-time attention	Worker is engaged 100% of the time in operating the equipment. The task can be (1) repetitive and cyclical or (2) nonrepetitive.	Worker operating a drop forge hammer (repetitive and cyclical). Worker on an assembly line whose task time equal to available service time (repetitive and cyclical). Truck driver driving an 18-wheeler (nonrepetitive). Worker mowing the lawn (nonrepetitive).
Part-time attention during each work cycle	Worker is engaged less than 100% of the time in operating the equipment. Task is repetitive and cyclical.	Worker loading and/or unloading production machine each cycle. Machine processes work units on mechanized or semiautomatic cycle. Worker is idle during machine cycle.
Periodic attention with regular servicing	Worker must service machine at regular intervals that are greater than one work cycle	Crane operator in steel mill moving molten steel ingots after each heat cycle. The operator must spend most of his time waiting for the next heat. Worker loading and/or unloading an automated production machine every 20 cycles. The machine operates on automatic cycle for the 20 cycles, but its storage capacity is limited to 20 work units.
Periodic attention with random servicing	Worker must service machine at random intervals that average more than one work cycle	Maintenance worker repairing production equipment when it malfunctions at random times. Firefighters responding to alarms that occur at random times.

- A metal trades worker operating a power buffer to buff the surface of a metal part
- A carpenter using a power saw to cut standard lengths of lumber
- A forklift driver moving pallet loads from the truck dock to the storage racks in a warehouse.

In these cases when the work cycle is repetitive but the cycle time is not constant, the analysis methods in Section 1.2 can be used.

Examples of worker–machine systems that operate on a nonrepetitive work cycle include the following:

- A trucker driving a tractor-trailer on an interstate highway
- A construction worker operating a backhoe
- A farmer operating a threshing machine to separate seeds from crop
- A carpenter using portable power tools to build a deck on a newly constructed house.

These nonrepetitive work situations do not consist of a regular work cycle that is repeated over and over. The time to accomplish the work depends on the skill and work ethic of

the persons performing the tasks. Estimates or historical records based on previous similar jobs are often used to determine how long the work should take to complete.

In this section, we focus attention on the second category of worker–machine system, in which machine time is constant and does not depend on operator control, and the work cycle is repetitive. Two cases are discussed: (1) cycle times with no overlap between worker and machine and (2) worker–machine systems with internal work elements.

Cycle Times with No Overlap Between Worker and Machine. In a worker–machine system, the work elements include one or more actions and/or operations performed by the machine. If there is no overlap in work elements between the worker and the machine, then the normal time for the cycle is simply the sum of their respective normal times:

$$T_n = T_{nw} + T_m \tag{6}$$

where T_{nw} = normal time for the worker-controlled portion of the cycle, min; and T_m = machine cycle time (assumed constant).

To determine the standard time for the cycle, a machine allowance is sometimes added to the machine time. If we include such an allowance factor in the standard time calculation, we have

$$T_{std} = T_{nw}(1 + A_{pfd}) + T_m(1 + A_m) \tag{7}$$

where T_{nw} = normal time of the worker, min; T_m = constant time for the machine cycle, min; A_m = machine allowance factor, used in the equation as a decimal fraction; and the other terms have the same meaning as before.

A typical value used by companies for the machine allowance factor is $A_m = 30\%$. This tends to help workers achieve higher worker efficiencies, which is especially important if the worker is paid on an incentive basis. Workers might prefer to work on an entirely manual cycle if the machine allowance were not provided. On the other hand, some companies do not see the need to use a machine allowance, in which case $A_m = 0$. An argument for $A_m = 0$ is that the worker is idle during the machine cycle, and so does not have to expend any effort during this portion of the work cycle. Other companies simply set the A_m value to be the same as A_{pfd}. The following examples show how the value of A_m affects the standard time and worker efficiency.

Example 8 Effect of Machine Allowance on Standard Time

In the operation of a worker–machine system, the work cycle consists of several manual work elements (operator-controlled) and one machine element performed under semiautomatic control. One workpiece is produced each cycle. The manual work elements total a normal time of 1.0 min and the semiautomatic machine cycle is a constant 2.0 min. The PFD allowance factor A_{pfd} is 15%. Determine the standard time using (a) $A_m = 0$ and (b) $A_m = 30\%$.

Solution: The normal time for the work cycle is the normal time for the worker-controlled elements plus the machine cycle time:

$$T_n = 1.0 + 2.0 = 3.0 \text{ min}$$

(a) With a machine allowance of 0, the standard time is calculated as

$$T_{std} = 1.0(1 + 0.15) + 2.0 = 3.15 \text{ min}$$

(b) With a machine allowance of 30%, the standard time is

$$T_{std} = 1.0(1 + 0.15) + 2.0(1 + 0.30) = 1.15 + 2.60 = 3.75 \text{ min}$$ ■

Example 9 Effect of Machine Allowance on Worker Efficiency

Based on the standard times computed in (a) and (b) of the previous example, determine the worker efficiencies for two cases if 150 units are produced in one 8-hour shift.

Solution: **(a)** If $T_{std} = 3.15$ min, the number of standard hours accomplished is

$$H_{std} = 150(3.15) = 472.5 \text{ min} = 7.875 \text{ hr}$$

$$\text{Worker efficiency } E_w = 7.875/8.0 = 0.984 = 98.4\%$$

(b) If $T_{std} = 3.75$ min, the number of standard hours accomplished is

$$H_{std} = 150(3.75) = 562.5 \text{ min} = 9.375 \text{ hr}$$

$$\text{Worker efficiency } E_w = 9.375/8.0 = 1.172 = 117.2\%$$ ■

When the work cycle includes a machine cycle and the machine time is a constant, then operator performance has no effect on this machine element. The only way the worker's pace can affect the work cycle time is during those elements that are operator-controlled. The operator may be idle during the machine element, as in Examples 8 and 9, unless the work sequence can be designed to include operator work elements that are performed while the machine is running.

Worker–Machine Systems with Internal Work Elements. In the operation of a worker–machine system, it is important to distinguish between the operator's work elements that are performed in sequence with the machine's work elements and those that are performed simultaneously with the machine elements. Operator elements that are performed sequentially are called ***external work elements*** while those that are performed simultaneously with the machine cycle are called ***internal work elements***. The distinction is important because it is desirable to construct the work cycle sequence so that as many of the operator elements as possible are performed as internal elements. This tends to minimize the cycle time, as illustrated by the following example.

Example 10 Internal Versus External Work Elements in Cycle Time Analysis

The work cycle in a worker–machine system consists of the elements and associated times given in the table below. All of the operator's work elements are external to the machine time. Can some of the worker's elements be made internal to the machine cycle, and if so, what is the expected cycle time for the operation?

Sequence	Work Element Description	Worker Time (min)	Machine Time (min)
1	Worker walks to tote pan containing raw stock	0.13	(idle)
2	Worker picks up raw workpart and transports to machine	0.23	(idle)
3	Worker loads part into machine and engages machine semiautomatic cycle	0.12	(idle)
4	Machine semiautomatic cycle	(idle)	0.75
5	Worker unloads finished part from machine	0.10	(idle)
6	Worker transports finished part and deposits into tote pan	0.15	(idle)
Totals		0.73	0.75

Solution: Since all of the work elements in the cycle are sequential, the total cycle time is the sum of the worker elements and the machine element:

$$T_c = 0.73 + 0.75 = 1.48 \text{ min.}$$

The worker is idle during the entire machine semiautomatic cycle. It should be possible to imbed elements 1, 2, and 6 as internal elements that are performed while the machine is running on a semiautomatic cycle. Using the times from the preceding table, the resulting work cycle can be organized as follows:

Sequence	Work Element Description	Worker Time (min)	Machine Time (min)
1	Worker unloads finished part from machine	0.10	(idle)
2	Worker loads part into machine and engages semiautomatic machine cycle	0.12	(idle)
3	Machine semiautomatic cycle		0.75
4	Worker transports finished part and deposits it into tote pan, walks to tote pan containing raw stock, and picks up raw workpart and transports it to machine. (This element is internal to the machine semiautomatic cycle.)	0.15 + 0.13 + 0.23 = 0.51	
Totals		0.73	0.75

Although the total times for the worker and the machine are the same as before, element 4 in the revised cycle (which consists of elements 1, 2, and 6 from the original work cycle) is performed simultaneously with the machine time, resulting in the following new cycle time:

$$T_c = 0.10 + 0.12 + 0.75 = 0.97 \text{ min}$$

This represents a 34% reduction in cycle time, which translates into a 53% increase in production rate.

■

When internal elements are present in the work cycle, it must then be determined whether the machine cycle time or the sum of the worker's internal elements take longer. To determine the normal time for the cycle,

$$T_n = T_{nw} + \text{Max}\{T_{nwi}, T_m\} \tag{8}$$

where T_{nw} = normal time for the worker's external elements, min; T_{nwi} = normal time for the worker's internal elements, min; and T_m = machine cycle time. The standard time for the cycle is given by

$$T_{std} = T_{nw}(1 + A_{pfd}) + \text{Max}\,\{T_{nwi}(1 + A_{pfd}),\, T_m(1 + A_m)\} \tag{9}$$

where A_{pfd} and A_m are the worker's allowance factor and the machine allowance factor, respectively. Finally, the actual cycle time depends on the worker's performance level, applied to the normalized times as

$$T_c = \frac{T_{nw}}{P_w} + \text{Max}\left(\frac{T_{nwi}}{P_w},\, T_m\right) \tag{10}$$

where P_w is the worker performance level during the cycle, expressed as a decimal fraction; and the other symbols mean the same as before. We assume in equation (10) that the worker's performance level is the same on the external and internal elements. If these P_w values are different, then the computations must reflect these differences.

3 AUTOMATED WORK SYSTEMS

Automation is the technology by which a process or procedure is accomplished without human assistance. It is implemented using a program of instructions combined with a control system that executes the instructions. Power is required to drive the process and to operate the program and control system.

There is not always a clear distinction between worker–machine systems and automated systems, because many worker–machine systems operate with some degree of automation. Let us distinguish between two forms of automation: semiautomated and fully automated. A ***semiautomated machine*** performs a portion of the work cycle under some form of program control, and a human worker tends to the machine for the remainder of the cycle, by loading and unloading it, or performing some other task during each cycle. An example of this category is an automated lathe controlled for most of the work cycle by the part program but requiring a worker to unload the finished part and load the next workpiece at the end of each machine cycle. In these cases, the worker must attend to the machine during every cycle. This type of operation has the same characteristics as a worker–machine system that requires the part-time attention of the worker during each work cycle (Table 3). Its cycle time analysis is discussed in Section 2.2.

The continuous presence of the operator during the cycle may not always be required. If the automatic machine cycle takes, say, 10 min while the part unloading and loading portion of the work cycle takes only 30 sec, then there may be an opportunity for one worker to tend more than one machine. We analyze this possibility in Section 5.

A ***fully automated machine*** is distinguished from its semiautomated cousin by the capacity to operate for extended periods of time with no human attention. By extended periods of time, we mean longer than one work cycle. A worker is not required to be present during each cycle. Instead, the worker may need to tend the machine every tenth cycle, or every hundredth cycle. An example of this type of operation is found in many injection molding plants, where the molding machines run on automatic cycle, but periodically the molded parts at the machine must be collected by a worker. This case is identified in Table 3 as periodic attention with regular servicing.

Certain fully automated processes require one or more workers to be present to continuously monitor the operation and make sure that it performs according to the intended specifications. Examples of these kinds of automated processes are found at chemical-processing facilities, oil refineries, and nuclear power plants. The workers do not actively participate in the process except to make occasional adjustments in the equipment settings, to perform periodic maintenance, and to spring into action if something goes wrong.

4 DETERMINING WORKER AND MACHINE REQUIREMENTS

One of the problems faced by any organization is determining the appropriate staffing levels. How many workers are required to achieve the organization's work objectives? If too few workers are assigned to perform a given amount of work, then the work cannot be completed on time, and customer service suffers. If too many workers are assigned, then payroll costs are higher than needed, and productivity suffers. Determining the number of workers or worker–machine systems that will be required to accomplish a specified amount of work is the problem we address in this section. The basic approach consists of two steps:

1. Determine the total workload that must be accomplished in a certain period (hour, week, month, year), where ***workload*** is defined as the total hours required to complete a given amount of work or to produce a given number of work units scheduled during the period.
2. Divide the workload by the available time per worker, where ***available time*** is defined as the number of hours in the same period available from one worker or worker–machine system.

Let us consider two general cases: (1) when setup time is not a factor and (2) when setup time must be included in the determination.

4.1 When Setup Is Not a Factor

Workload is figured as the quantity of work units to be produced during the period of interest multiplied by the time (hours) required for each work unit. The time required for each work unit is the work cycle time in most cases, so that workload is given by

$$WL = QT_c \tag{11}$$

when several machines require servicing simultaneously, thus overloading the capabilities of the human worker. In addition, at other times during the shift, the worker will have no machines to service and will therefore be idle.

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REVIEW QUESTIONS

1 In terms of human participation, what are the three basic categories of work systems?

2 What is the general characteristic that is common to nearly all pure manual work?

3 What is the one best method principle?

4 What is meant by the term *normal performance*?

5 What is meant by the term *normal time* for a task?

6 What does PFD stand for? What is the purpose of the PFD allowance in determining the standard time for a task?

7 What is an irregular work element?

8 Define the meaning of worker efficiency.

9 What is a worker–machine system?

10 What are the three main categories of powered machinery in worker–machine systems?

11 Define machine tool.

12 Cycle times in worker–machine systems divide into two categories: (1) machine time depends on operator and (2) machine time is constant and repetitive. Give an example of each category.

13 What is the difference between an external work element and an internal work element in a worker–machine cycle?

14 What are the factors that affect the workload calculation when determining worker requirements?

15 What does availability mean?

16 What is a machine cluster?

PROBLEMS

Cycle Time Analysis of Manual Work

1 If the normal time is 1.30 min for a repetitive task that produces one work unit per cycle, and the company uses a PFD allowance factor of 12%, determine (a) the standard time for the task and (b) how many work units are produced in an 8-hour shift at standard performance.

2 The normal time for a repetitive task that produces two work units per cycle is 3.0 min. The plant uses a PFD allowance factor of 15%. Determine (a) the standard time per piece and (b) how many work units are produced in an 8-hour shift at standard performance.

3 The normal time to perform a certain manual work cycle is 3.47 min. In addition, an irregular work element whose normal time is 3.70 min must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%. Determine (a) the standard time per piece and (b) how many work units are produced during an 8-hour shift at 100% performance, and the worker works exactly 7.018 hr, which corresponds to the 14% allowance factor. (c) If the worker's pace is 120% and he works 7.2 hours during the regular shift, how many units are produced?

4 The normal time to perform a repetitive manual assembly task is 4.25 min. In addition, an irregular work element whose normal time is 1.75 min must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance factor is 16%. Determine (a) the standard time per piece and (b) how many work units are produced in an 8-hour shift at standard performance. (c) Determine the anticipated amount of time worked and the amount of time lost per 8-hour shift that corresponds to the PFD allowance factor of 16%.

5 The standard time for a manual material-handling work cycle is 2.58 min per piece. The PFD allowance factor used to set the standard was 13%. During a particular 8-hour shift of interest, it is known that the worker lost a total of 53 min due to personal time, rest breaks, and delays. On that same day, the worker completed 214 work units. Determine (a) the number of standard hours accomplished, (b) worker efficiency, and (c) the worker's performance level expressed as a percentage.

6 A worker performs a repetitive assembly task at a workbench to assemble products. Each product consists of 25 components. Various hand tools are used in the task. The standard time for the work cycle is 7.45 min, based on using a PFD allowance factor of 15%. If the worker completes 75 product units during an 8-hour shift, determine (a) the number of standard hours accomplished and (b) worker efficiency. (c) If the worker took only one rest break, lasting 13 min, and experienced no other interruptions during the 8 hours of shift time, determine her worker performance.

Cycle Time Analysis in Worker–Machine Systems

7 The normal time of the work cycle in a worker–machine system is 5.39 min. The operator-controlled portion of the cycle is 0.84 min. One work unit is produced each cycle. The machine cycle time is constant. (a) Using a PFD allowance factor of 16% and a machine allowance factor of 30%, determine the standard time for the work cycle. (b) If a worker assigned to this task completes 85 units during an 8-hour shift, what is the worker's efficiency? (c) If it is known that a total of 42 min was lost during the 8-hour clock time due to personal needs and delays, what was the worker's performance on the portion of the cycle he controlled?

8 A worker is responsible for loading and unloading a production machine. The load/unload elements in the repetitive work cycle have a normal time of only 24 sec, and the machine cycle time is 2.83 min. One part is produced each cycle. Every sixth cycle, the operator must replace the tote pans of parts, which takes 2.40 min (normal time). For setting the standard time, the PFD allowance factor is 15% and the machine allowance factor is 15%. Determine the standard time under the following alternative assumptions: (a) the irregular element is performed as an external element and (b) the irregular element is performed as an internal

element. (c) Determine the corresponding standard daily production quantities (8-hour shift) for each of these time standards.

9 The work cycle in a worker–machine system consists of (1) external manual work elements with a total normal time of 0.42 min, (2) a machine cycle with a machine time of 1.12 min, and (3) internal manual elements with a total normal time of 1.04 min. (a) Determine the standard time for the cycle, using a PFD allowance factor of 15% and a machine allowance factor of 30%. (b) How many work units are produced daily (8-hour shift) at standard performance?

10 Solve the previous problem but assume that the machine allowance factor is 0%.

11 The normal time for a work cycle in a worker–machine system is 6.27 min. For setting the standard time, the PFD allowance factor is 12% and the machine allowance factor is 25%. The work cycle includes manual elements totaling a normal time of 5.92 min, all but 0.65 min of which are performed as internal elements. Determine (a) the standard time for the cycle and (b) the daily output at standard performance. (c) During an 8-hour shift, the worker lost 39 min due to personal time, rest breaks, and delays, and she produced 72 pieces. What was the worker's pace on the operator-controlled portion of the shift?

12 Solve the previous problem but assume that the machine allowance factor is 0%.

Determining Worker and Work Cell Requirements

13 A total of 1000 units of a certain product must be completed by the end of the current week. It is now late Monday afternoon, so only four days (8-hour shifts) are left. The standard time for producing each unit of the product (all manual operations) is 11.65 min. How many workers will be required to complete this production order if it is assumed that worker efficiency will be 115%?

14 Future production requirements in the turret lathe department must be satisfied through the acquisition of several new machines and the hiring of new operators, the exact number to be determined. There are three new parts that will be produced. Part A has annual quantities of 20,000 units; part B, 32,000 units; and part C, 47,000 units. Corresponding standard times for these parts are 7.3 min, 4.9 min, and 8.4 min, respectively. The department will operate one 8-hour shift for 250 days/yr. The machines are expected to be 98% reliable, and the anticipated scrap rate is 4%. Worker efficiency is expected to be 100%. How many new turret lathes and operators are required to meet these production requirements?

15 A new stamping plant must supply an automotive final assembly plant with stampings, and the number of new stamping presses must be determined. Each press will be operated by one worker. The plant will operate one 8-hour shift per day, five days per week, 50 weeks per year. The plant must produce a total of 20,000,000 stampings annually. However, 400 different stamping designs are required, in batch sizes of 5000 each, so each batch will be produced 10 times per year to minimize build-up of inventory. Each stamping takes 6 sec on average to produce. Scrap rate averages 2% in this type of production. Before each batch, the press must be set up, with a standard time per setup of 3.0 hours. Presses are 95% reliable (availability = 95%) during production and 100% reliable during setup. Worker efficiency is expected to be 100%. How many new stamping presses and operators will be required?

16 Solve the previous problem, except the plant will operate two 8-hour shifts instead of one. (a) How much money would be saved if each press has an investment and installation cost of $250,000. (b) If each worker's wage rate is $15.00/hr, how much money would be saved by operating two 8-hour shifts per day rather than one 8-hour shift?